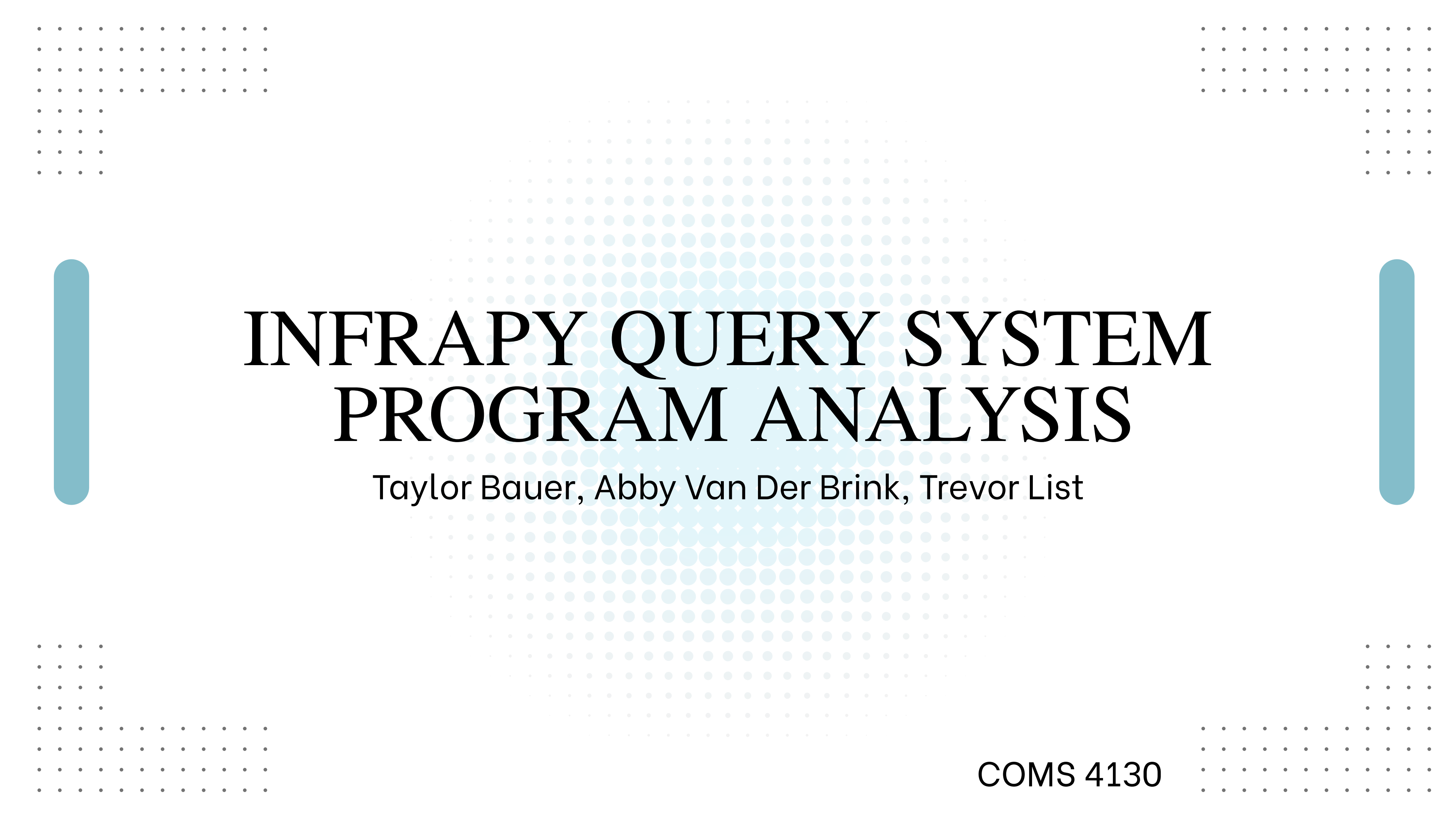




INFRAPY QUERY SYSTEM PROGRAM ANALYSIS

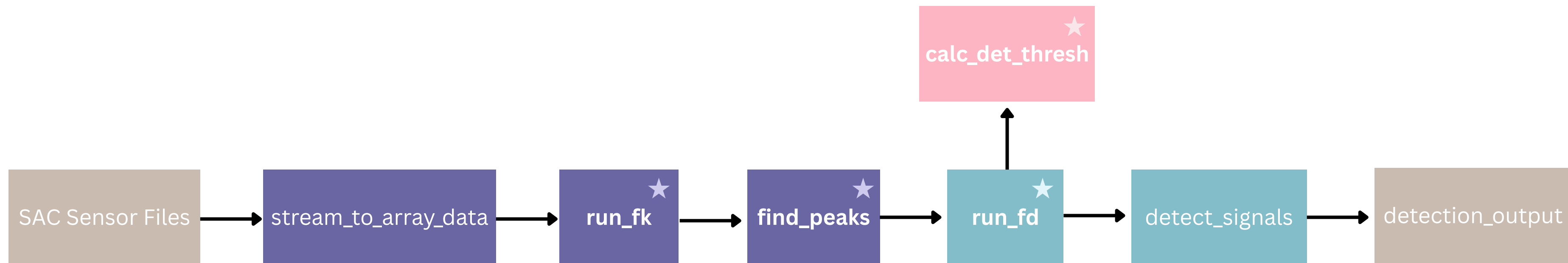
Taylor Bauer, Abby Van Der Brink, Trevor List

BACKGROUND

- Python program developed by Los Alamos National Lab (LANL) for infrasonic soundwave detection in nuclear treaty monitoring
- Uses Adaptive Fisher Detector algo (AFD) which compares signals across recordings from 4 ground sensors and marks a detection when signal strength exceeds a threshold that adjusts based on background noise
- `beamforming_new.py`

OUR PIPELINE FOCUS

(1) Raw sensor recordings are loaded & processed into an array (2) scanned in short time windows to find signal direction and strength and strength (3) Detector decides if a real detection happened or not



Target functions (focus on detection and signal processing)

- **run_fk**: signal processing entry point, processes sensor recordings in the time windows
- **find_peaks**: finds the strongest signal direction from each time window and sends to detector
- **run_fd**: (AFD) where the actual detection decision happens! decides if a signal is strong and consistent enough to count as a detection
- **calc_det_thresh**: threshold computed for detection events based on background noise level

QUERIES & MOTIVATION (1)

Queries

Call Graph

- find what functions call other functions
- find functions that are called by a given function
- call path between two functions?

Dataflow

- show where variables are used, or passed between functions
- find if two variables are data dependent

Motivations (General)

understand how functions depend on each other & the structure of the program

understand how variables move through the program

QUERIES & MOTIVATION (2)

Queries

Coverage

- find coverage of a file
- find overall coverage summary
- find what lines were never executed

Fuzzing

- find runs that passed when using invalid input
- find runs that failed & why
- fuzzing runs for files

Motivations (General)

know how much of the program was tested and find branches that could have bugs

test how the program handles unexpected inputs and find inputs that cause crashes or other weird behavior

INTERNAL WORKFLOW

Queries, Tools, & Output

Call Graph (pycallgraph2)
call.json

Dataflow (ast module)
df.json

Coverage (trace module)
covg.json

Fuzzing (hypothesis)
fuzz_results.json

SQLite DB (4130db.db)
4 tables

Query System

Modules

call_graph_query.py

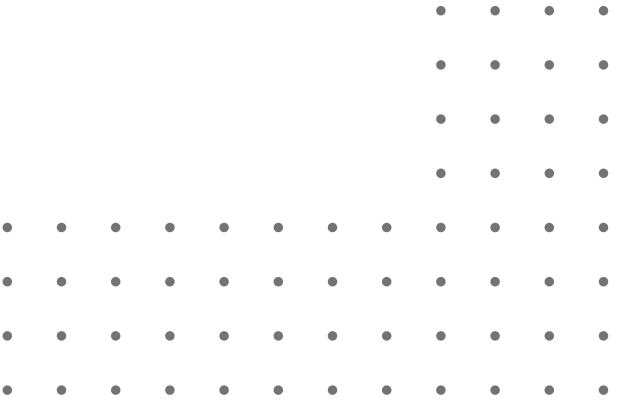
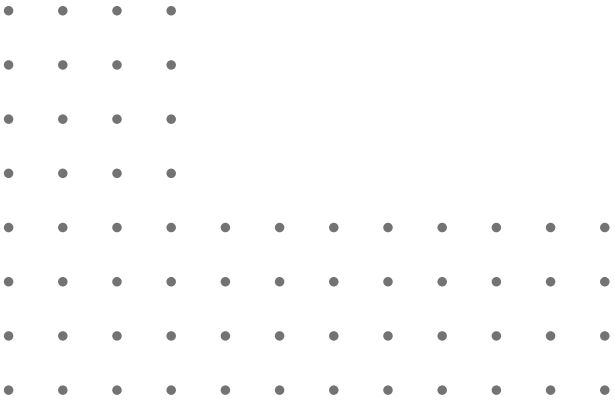
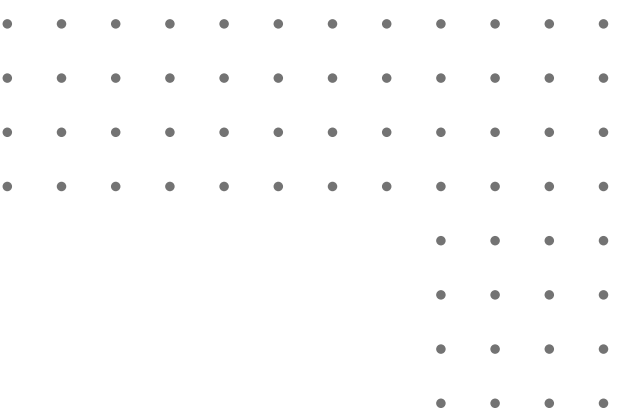
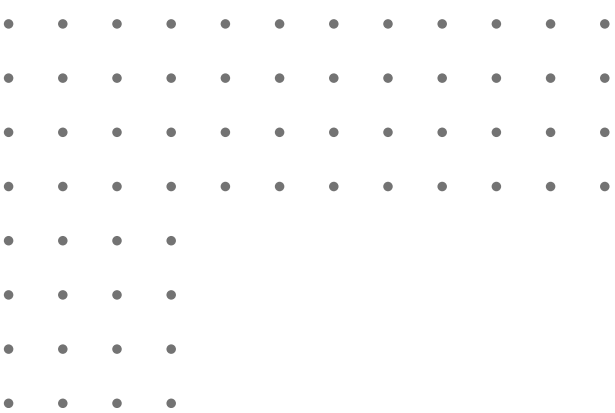
coverage_query.py

fuzzing_query.py

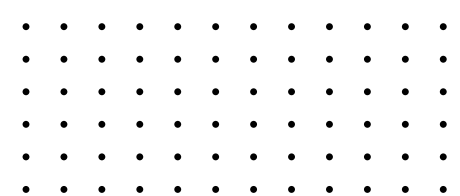
natural language router

Interactive Shell/GUI

**modules combined in query router*



DEMO



REFERENCES



- Jeremy D. Webster, philblomand SpacelySpaceSprockets, “LANL-Seismoacoustics/infrapy: v0.4”. Zenodo, Sep. 27, 2022. doi: 10.5281/zenodo.7116991.
- Infrapy Link (Repository): <https://github.com/LANL-Seismoacoustics/infrapy>
- Pycallgraph2 Docs (CFG): <https://pypi.org/project/pycallgraph2/>
- AST Docs (Dataflow): <https://docs.python.org/3/library/ast.html>
- Trace Docs (Coverage): <https://docs.python.org/3/library/trace.html>
- Hypothesis Docs (Fuzzing): <https://hypothesis.readthedocs.io/en/latest/>